

Investing aflevering 2

Question1

Consider a standard binomial model for the price of a non-dividend-paying stock, S_t , which starts at $S_0 = 50$ and in each 1/4-year period goes up by a multiplicative factor $u = 1.1$ or down by a multiplicative factor $d = 0.95$. In the entire tree, it is equally likely that the stock will rise or fall ($p = 1 - p = 0.5$) over the next period. Further assume that the risk-free interest rate is zero.

- a) Is the model arbitrage free and complete

We write the state price matrix after 1 period with the given information (we just adjust for 1 dollar in the underlying)

$$A = \begin{pmatrix} d & u \\ 1 + r_f & 1 + r_f \end{pmatrix} = \begin{pmatrix} 0.95 & 1.1 \\ 1 & 1 \end{pmatrix}$$

$$\psi = A^{-1}P = \begin{pmatrix} 0.95 & 1.1 \\ 1 & 1 \end{pmatrix}^{-1} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \approx \begin{pmatrix} 0.6667 \\ 0.3333 \end{pmatrix} > 0$$

The matrix is invertible and the stateprices are larger than 0 thus the market is complete and free of arbitrage

(we also have a “theorem” that says that a binomial model of stock prices with fixed up and downs is arbitragefree and complete if and only if $d < \frac{1}{d_{0,t}} < u$)

- b) Calculate $\mathbb{E}^P \left[\frac{S_1}{S_0} - 1 \right]$

$$\mathbb{E}^P \left[\frac{S_1}{S_0} - 1 \right] = \frac{\mathbb{E}^P[S_1]}{S_0} - 1 = (0.5 \cdot 0.95 + 0.5 \cdot 1.1) - 1 \approx 0.025$$

- c) Find the risk neutral probabilities

$$q = \frac{1 - d}{u - d} = \frac{1 - 0.95}{1.1 - 0.95} \approx 0.3333$$

$$1 - q = \frac{2}{3}$$

- d) Calculate $\mathbb{E}^Q \left[\frac{S_1}{S_0} - 1 \right]$ and interperate the result

$$\mathbb{E}^Q \left[\frac{S_1}{S_0} - 1 \right] = \frac{\mathbb{E}^Q[S_1]}{S_0} - 1 = \left(\frac{2}{3} \cdot 0.95 + \frac{1}{3} \cdot 1.1 \right) - 1 = 0$$

The risk neutral measure is exactly the measure where our expected return is equal to the risk free one since our risk free rate is 0 we expect our expected return to be the same

- e) Vi regner

$$E^Q[C_0] = q^4 \cdot c_{u^4} + 4 \cdot q^3 \cdot (1 - q) \cdot c_{u^3d} + 6 \cdot q^2(1 - q)^2 \cdot c_{u^2d^2} + 4 \cdot q(1 - q)^3 \cdot c_{ud^3} + (1 - q)^4 \cdot c_{d^4}$$

$$= \frac{1}{3} \cdot 23.205 + 4 \cdot \frac{1}{3^3} \cdot \left(\frac{2}{3} \right)^1 \cdot 13.225 + 6 \cdot \frac{1}{3^2} \cdot \left(\frac{2}{3} \right)^2 \cdot 4.60125 \approx 2.956$$

- f) S is in a state equivalent with up up down

Thus we must have that the price either goes to 2up 2down or 3up 1 down

Thus the value of the option either goes to c_{u^3d} or $c_{u^2d^2}$

Hedging in the underlying and the bankaccount must have 0 cashflows at time 1

Thus

$$\Delta \cdot 63.2225 + B = 13.2225$$

$$\Delta \cdot 54.60125 + B = 4.60125$$



Ligningssystemet løses for Δ, B vha. WordMat's 'Løs Ligninger' funktion,

$$\Delta = 1 \quad \wedge \quad B = -50$$

thus to hedge the short position in the option we must be long 1 underlying and loan 50

- g) What is the price of an American call with strike 50?

Since no dividends

$$c_A = C_E = 2.956$$

- h) Price an American put with $K=70$

We calculate it backwards in each step deciding if we want to exercise i.e the value at each node is exactly $\max\{K - s_t, P_t, 0\}$ where the 0 is irrelevant since $P_t \geq 0$

This pricing gives us the tree

					0
			4,518333		
		9,856111		6,7775	
	15,1187		12,525		
20,03957		17,75		15,39875	
	22,5		20,3625		
		24,875		22,84438	
			27,13125		
				29,27469	

Pricing the put price at 20.04

We also see that we basically always want to exercise early unless we hit up all the times

- i) $u = e^{\mu \cdot \Delta t + \sigma \cdot \sqrt{\Delta t}}, d = e^{\mu \cdot \Delta t - \sigma \cdot \sqrt{\Delta t}}$ find μ and σ

$$\Delta t = \frac{1}{4}, u = 1.1, d = 0.95$$

$$u \cdot d = e^{2 \cdot \mu \cdot \Delta t}$$

$$1.1 \cdot 0.95 = e^{2 \cdot \mu \cdot \frac{1}{4}}$$



Ligningen løses for μ vha. WordMat.

$$\mu = 0,08803$$

$$1.1 = e^{0,08803 \cdot \frac{1}{4} + \sigma \cdot \frac{1}{2}}$$



Ligningen løses for σ vha. WordMat.

$$\sigma = 0,1466$$

Question 2

Answer all subquestions!

a. Let W_t be a standard Brownian motion. Derive the dynamics of $f(W_t) = W_t^2$ and determine if W_t^2 is a martingale.

b. Let S_t be a Geometric Brownian motion with dynamics

$$dS_t = \mu S_t dt + \sigma S_t dW_t.$$

Assume that $X_t = S_t^2$. Determine the dynamics of X_t , that is, find dX_t . Explain your findings.

Consider a *power call* option: It has an expiry T , strike price K , and the payoff is

$$\frac{1}{2} \max \left\{ \frac{(S_T)^2}{K} - K, 0 \right\}$$

at expiry T .

a) Using Itô's lemma

$$\begin{aligned} df &= \left(\frac{\partial f}{\partial t} + \frac{\partial f}{\partial S} \cdot 0 + \frac{1}{2} \cdot \frac{\partial^2 f}{\partial S^2} \right) dt + \frac{\partial f}{\partial S} dW_t \\ &= dt + 2W_t dW_t \end{aligned}$$

If the process is a martingale then the trend must be 0

Since the trend is not 0 the process is not a martingale

b) $dS_t = \mu \cdot S_t dt + \sigma \cdot S_t dW_t$, $X_t = S_t^2$

Again Itô's lemma

$$\begin{aligned} dX_t &= \left(\frac{\partial X}{\partial t} + \frac{\partial X}{\partial S} \cdot S_t \cdot \mu + \frac{1}{2} \cdot \frac{\partial^2 X}{\partial S^2} \right) dt + \frac{\partial X}{\partial S} dW_t \\ &= \left(\frac{\partial X}{\partial S} \cdot S_t \cdot \mu + \frac{1}{2} \cdot \frac{\partial^2 X}{\partial S^2} \cdot \sigma^2 \cdot S_t^2 \right) dt + S_t \cdot \sigma \cdot \frac{\partial X}{\partial S} dW_t \\ &= (2 \cdot \mu \cdot S_t^2 + \sigma^2 S_t^2) dt + 2 \cdot S_t^2 \cdot \sigma dW_t \end{aligned}$$

Which can be redefined as a geometric brownian motion

$$= X_t(2\mu + \sigma^2)dt + 2\sigma \cdot X_t dW_t$$

This is consistent with an exercise we've done for an arbitrary exponent

c. Show that the price of a *power call* is always higher than the price of a European call with the same strike price and expiry.

Hint: Consider the payoff functions of the two options.

d. Express X_T using X_0 , model parameters, and the value of the Wiener process at T , W_T .

e. Write down what SDE the price of the power call option must satisfy.

f. Determine the price of the power call either by risk neutral pricing or by the SDE-solution approach. Does it satisfy the property identified in part c)?

Hint: You can make use of the results in parts d) or e) and refer to the Black-Scholes formula derived in the lectures.

c) Assume for contradiction $C_P < C_E$

Now look at the cashflows of selling a european and buying a Power, paying you

$$C_E - C_P > 0$$

At time T

If $K > S_T$

$$\frac{1}{2} \max \left\{ \frac{S^2}{K} - K, 0 \right\} - \max \{ S - K, 0 \} = 0$$

if $K \leq S_T$

$$\begin{aligned} \frac{1}{2} \max \left\{ \frac{S^2}{K} - K, 0 \right\} - \max \{ S - K, 0 \} &= \frac{1}{2} \left(\frac{S^2}{K} - K \right) - (S - K) \\ &= \frac{1}{2} \left(\frac{S^2}{K} - 2S + K \right) = \frac{1}{2 \cdot K} (S - K)^2 \geq 0 \end{aligned}$$

Thus we have a contradiction since no arbitrage states we cant only have negative or positive payoffs thus we must have negative cashflow at time 0

$$C_E - C_P \leq 0$$

$$C_E \leq C_P$$

d) We take the log process

$$\begin{aligned} d \log(X_T) &= \left(\frac{1}{X_T} \cdot X_T (2\mu + \sigma^2) + \frac{1}{2} \left(-\frac{1}{X_t^2} \cdot X_t^2 \right) \cdot 4 \cdot \sigma^2 \right) dt + 2\sigma dW_t \\ &= ((2\mu - \sigma^2))dt + 2\sigma dW_t \end{aligned}$$

Now we write up the equivalent statement

$$\begin{aligned} \log(X_T) &= \log(X_0) + T \cdot (2\mu - \sigma^2) + 2\sigma W_T \\ X_T &= X_0 \cdot e^{T(2\mu - \sigma^2) + 2\sigma W_T} = X_0 \cdot e^{T(2\mu - \sigma^2) + 2\sigma \sqrt{T} \cdot W} \end{aligned}$$

Where W is standard normal

e) The Power calls price process must satisfy the BSM equation since we are in a BSM world

$$dC_t^P = \left(\frac{\partial C_t^P}{\partial t} + \frac{\partial C_t^P}{\partial S} \cdot S_t \cdot \mu + \frac{1}{2} \frac{\partial^2 C_t^P}{\partial S^2} \cdot \sigma^2 \cdot S_t^2 \right) dt + \frac{\partial C_t^P}{\partial S} \cdot S_t \cdot \sigma dW_t$$

A no arbitrage argument also implies

$$rC^P = \frac{\partial C_t^P}{\partial t} + \frac{\partial C_t^P}{\partial S} \cdot S_t \cdot \mu + \frac{1}{2} \frac{\partial^2 C_t^P}{\partial S^2} \cdot \sigma^2 \cdot S_t^2$$

Which is the black scholes equation

f) Under the risk neutral measure

Girsanovs theorem states if

$$dS_t = \mu \cdot S_t dt + \sigma \cdot S_t dW_t$$

Then $dW_t^Q = dW_t + \frac{\mu - r}{\sigma}$ thus

$$dS_t = \mu \cdot S_t dt + \sigma \cdot S_t dW_t = r \cdot S_t dt + \sigma \cdot S_t dW_t^Q$$

We have been given a formula for pricing under the risk neutral measure in the BSM world

It just involves a nasty integral

$$\begin{aligned} \mathbb{E}^Q \left[\frac{1}{2} \max \left\{ \frac{S^2}{K} - K, 0 \right\} \right] \\ = \int_{-\infty}^{\infty} \frac{1}{2} \max \left\{ \frac{S^2}{K} - K, 0 \right\} \cdot \frac{1}{S} \cdot \frac{1}{\sqrt{2 \cdot \pi \cdot T}} \cdot e^{\frac{-1}{2} \left(\log(S) - \log(S_0) - \left(r - \frac{\sigma^2}{2} \right) T \right)} \cdot \frac{1}{\sigma^2 \cdot T} ds \end{aligned}$$

Now we just need to "holde tungen lige i munden og løse den her satan" as we say in danish

The integral is 0 for $S < K$ so

$$= \frac{1}{2} \int_K^\infty \left(\frac{S^2}{K} - K \right) \cdot \frac{1}{S} \cdot \frac{1}{\sqrt{2 \cdot \pi \cdot \sigma^2 \cdot T}} \cdot e^{\frac{-\frac{1}{2}(\log(S) - \log(s_0) - (r - \frac{\sigma^2}{2})T)}{\sigma^2 \cdot T}} ds$$

We swap variables

$$\begin{aligned} y &= \log(s) \\ \frac{dy}{ds} &= \frac{1}{s} \Leftrightarrow ds = dy \cdot s \end{aligned}$$

And adjust the integral also note that $S = e^y$

$$= \frac{1}{2} \int_{\log(K)}^\infty \left(\frac{e^{2 \cdot y}}{K} - K \right) \cdot \frac{1}{\sqrt{2 \cdot \pi \cdot \sigma^2 \cdot T}} \cdot e^{\frac{-\frac{1}{2}(y - \log(s_0) - (r - \frac{\sigma^2}{2})T)}{\sigma^2 \cdot T}} dy$$

We split it up

$$\begin{aligned} &= \frac{1}{2} \int_{\log(K)}^\infty \frac{e^{2 \cdot y}}{K} \cdot \frac{1}{\sqrt{2 \cdot \pi \cdot \sigma^2 \cdot T}} \cdot e^{\frac{-\frac{1}{2}(y - \log(s_0) - (r - \frac{\sigma^2}{2})T)}{\sigma^2 \cdot T}} dy \\ &\quad - \frac{1}{2} \int_{\log(K)}^\infty K \cdot \frac{1}{\sqrt{2 \cdot \pi \cdot \sigma^2 \cdot T}} \cdot e^{\frac{-\frac{1}{2}(y - \log(s_0) - (r - \frac{\sigma^2}{2})T)}{\sigma^2 \cdot T}} dy \end{aligned}$$

We swap variables

$$\begin{aligned} \frac{(y - \log(s_0) - (r - \frac{\sigma^2}{2})T)^2}{\sigma^2 \cdot T} &= x^2 \Rightarrow x = \frac{y - \log(s_0) - (r - \frac{\sigma^2}{2})T}{\sigma \cdot \sqrt{T}} \\ \frac{dx}{dy} &= \frac{1}{\sigma \cdot \sqrt{T}} \Leftrightarrow dy = dx \cdot \sigma \cdot \sqrt{T} \end{aligned}$$

We also note that this is function taken on $\log(K)$ is exactly $-d_2$ from the BSM equation for standard call and put

Also:

$$\begin{aligned} x &= \frac{y - \log(s_0) - (r - \frac{\sigma^2}{2})T}{\sigma \cdot \sqrt{T}} \\ y &= \frac{2 \cdot \sqrt{T} \cdot \sigma \cdot x - T \cdot \sigma^2 + 2 \cdot \log(s_0) + 2 \cdot T \cdot r}{2} \end{aligned}$$

These imply

$$\begin{aligned} &= \frac{1}{2} \int_{-d_2}^\infty \frac{e^{2 \cdot \frac{2 \cdot \sqrt{T} \cdot \sigma \cdot x - T \cdot \sigma^2 + 2 \cdot \log(s_0) + 2 \cdot T \cdot r}{2}}}{K} \cdot \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{x^2}{2}} dx \\ &\quad - \frac{1}{2} \int_{-d_2}^\infty K \cdot \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{x^2}{2}} dx \end{aligned}$$

The second integral is exactly the standard normal CDF thus

$$\begin{aligned} -\frac{1}{2} \int_{-d_2}^\infty K \cdot \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{x^2}{2}} dx &= -\frac{K}{2} \int_{-d_2}^\infty \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{x^2}{2}} dx = -\frac{K}{2} (1 - \int_{-\infty}^{-d_2} \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{x^2}{2}} dx) \\ &= -\frac{K}{2} N(d_2) \end{aligned}$$

Now to the harder part

$$\begin{aligned}
&= \frac{1}{2} \int_{-d_2}^{\infty} \frac{e^{2 \cdot \sqrt{T} \cdot \sigma \cdot x - T \cdot \sigma^2 + 2 \cdot \log(s_0) + 2 \cdot T \cdot r}}{K} \cdot \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{x^2}{2}} dx \\
&= \frac{1}{2K} \cdot \int_{-d_2}^{\infty} e^{2 \cdot \sqrt{T} \cdot \sigma \cdot x - T \cdot \sigma^2 + 2 \cdot \log(s_0) + 2 \cdot T \cdot r} \cdot \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{x^2}{2}} dx \\
&= e^{2rT} \cdot \frac{S^2}{2K} \cdot \int_{-d_2}^{\infty} e^{2 \cdot \sqrt{T} \cdot \sigma \cdot x - T \cdot \sigma^2} \cdot \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{x^2}{2}} dx
\end{aligned}$$

We wish for a way to make it standard normal again we get the great idea to make a square

$$\frac{2 \cdot \sqrt{T} \cdot \sigma \cdot x - T \cdot \sigma^2 - \frac{x^2}{2}}{2}$$

$$\begin{aligned}
&\frac{4 \cdot \sqrt{T} \cdot \sigma \cdot x - 2T \cdot \sigma^2 - x^2}{2} - T\sigma^2 = -\frac{(x - 2\sigma\sqrt{T})^2}{2} + T\sigma^2 \\
&e^{2rT} \cdot \frac{S^2}{2K} \cdot \int_{-d_2}^{\infty} e^{2 \cdot \sqrt{T} \cdot \sigma \cdot x - T \cdot \sigma^2} \cdot \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{x^2}{2}} dx \\
&= e^{2rT} \cdot \frac{S^2}{2K} \cdot \int_{-d_2}^{\infty} e^{T \cdot \sigma^2} \cdot \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{(x - 2\sigma\sqrt{T})^2}{2}} dx
\end{aligned}$$

We change variables again

$$\begin{aligned}
&z = x - 2\sigma\sqrt{T} \\
&\frac{dz}{dx} = 1 \\
&= e^{2rT} \cdot \frac{S^2}{2K} \cdot e^{T \cdot \sigma^2} \cdot \int_{-d_2 - 2\sigma\sqrt{T}}^{\infty} \frac{1}{\sqrt{2 \cdot \pi}} \cdot e^{-\frac{z^2}{2}} dz = e^{(2r + \sigma^2)T} \cdot \frac{S^2}{2K} \cdot N(d_2 + 2\sigma\sqrt{T})
\end{aligned}$$

The price of a general derivative is given as

$$F_0 = \mathbb{E}^Q[d_{0,T} \cdot F_T]$$

Since continuous compounding in the BSM world

$$\begin{aligned}
C_0^P &= \mathbb{E}^Q[d_{0,T} \cdot C_T^P] = e^{-rT} \cdot [e^{(2r + \sigma^2)T} \cdot \frac{S^2}{2K} \cdot N(d_2 + 2\sigma\sqrt{T}) - \frac{K}{2} \cdot N(d_2)] \\
&= e^{(r + \sigma^2)T} \cdot \frac{S^2}{2K} \cdot N(d_2 + 2\sigma\sqrt{T}) - \frac{K}{2} N(d_2) \cdot e^{-rT}
\end{aligned}$$

Price must be higher since

$$C^P - C^E = e^{-rt} \cdot E^Q[(S - K)^2 \cdot \mathbb{I}_{S > K}] \geq 0$$

I've tried to compare prices directly but can't find a nice way to express it neatly

I suppose we could calculate this expectation under Q but I'd rather not